

Title

Patient perspectives and implementation factors shaping acceptance and sustained use of digital TB adherence support tools

Abstract

Background: Digital health tools for TB adherence require sustained patient engagement for effectiveness. Understanding facilitators and barriers to technology adoption in diverse, resource-limited settings is critical for implementation success. **Objective:** Qualitative analysis of patient experiences and perceptions regarding TB-TST implementation. **Methods:** Semi-structured interviews with 33 intervention-arm participants and 5 treatment support workers were conducted during or shortly after TB treatment completion. Data were analyzed using thematic coding informed by normalization process theory to identify factors enabling or hindering technology integration into routine TB care. **Results:** Patients identified app medication reminders and direct messaging with treatment supporters as most valuable; many used the app to ask sensitive questions avoided in face-to-face settings. Perceived benefits included improved organization, reduced treatment burden perceptions, and psychological reassurance. Initial confusion about the metabolite test purpose resolved over time through repeated explanation. Key barriers included inconsistent smartphone access, limited internet connectivity in some neighborhoods, burden of cumulative TB medication side effects, and work schedule conflicts with testing appointments. Qualitative feedback emphasized the importance of the treatment supporter relationship as a trusted anchor for technology use. **Conclusions:** Successful digital TB adherence tools require attention to social context, patient-centered design, and integration with human support systems rather than technology-only approaches.